

1. Title Page

Language Inequality in Large Language Models: Quantifying the Performance Gap in Low-Resource Turkic Languages (English, Russian, Kazakh, Uzbek)

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2. Abstract

Large language models (LLMs) have become de facto infrastructure for accessing information, yet the quality of that infrastructure is distributed highly unevenly across languages. This study empirically quantifies the magnitude of this gap across four languages occupying different positions on the resource spectrum: English, Russian, Kazakh, and Uzbek. Using a sample of 720 observations (60 prompts across three task categories — factual knowledge, logical reasoning, and translation — in each of four languages, run through three language models: llama-3.1-8b-instant, llama-3.3-70b-versatile, and openai/gpt-oss-20b), the results show that response accuracy does not decline monotonically with language resourcedness but instead drops sharply beyond a critical threshold: Russian is statistically indistinguishable from English (97.2% versus 96.7% accuracy), whereas both Turkic languages exhibit a comparable and substantial shortfall (86.1% and 85.0%, respectively). Contrary to the initial hypothesis, the most vulnerable task category proved to be not translation but logical reasoning. A prompt-level logistic regression ($n = 720$) confirms a statistically significant association between language resourcedness (number of Wikipedia articles) and the probability of a correct response ($OR = 4.32$ per unit of $\log_{10}$, $p < 0.0001$), controlling for task type and model. This work contributes a quantitative, reproducible measurement of LLM language inequality as it applies to the Turkic languages of Central Asia.

3. Research Objective

Large language models (LLMs) have de facto become infrastructure for accessing information: they are used for education, search, text generation, and decision support. Yet the quality of this infrastructure is distributed highly unevenly across languages. This study aims to empirically measure the magnitude of this gap for four languages representing different positions on the resourcedness scale: English (high-resource, dominant in training corpora), Russian (mid-resource, one of the largest languages in Common Crawl and Wikipedia), and Kazakh and Uzbek (low-resource Turkic languages of Central Asia, represented on the web at orders of magnitude lower scale).

The specific research task is not merely to confirm the already well-documented observation that “models perform worse on low-resource languages,” but to quantitatively estimate the magnitude of quality degradation by task type (factual questions, logical reasoning, translation) and to relate that degradation to objective metrics of language resourcedness. This allows the analysis to move from a qualitative observation to a measurable index of language inequality.

4. Research Hypothesis

H1 (primary). LLM response accuracy declines monotonically as language resourcedness decreases along the sequence English → Russian → Kazakh → Uzbek, with the degradation being non-linear: the gap between Russian and Kazakh/Uzbek is substantially larger than the gap between English and Russian, since the latter two languages are separated from the former by a “critical mass” threshold of pretraining data.

H2. The magnitude of degradation depends on task type: translation tasks will suffer the most (since they require high-quality generation in the target language), logic/reasoning tasks will suffer moderately, and factual questions will prove most sensitive to culturally specific gaps in the training data.

H3. There is a statistically significant positive correlation between a model's accuracy in a given language and the logarithm of the volume of that language's textual data in training corpora (proxied by the size of the corresponding Wikipedia edition), consistent with the notion of a “digital language divide” as a structural characteristic of contemporary LLMs.

5. Research Stages

The study was conducted in five sequential stages:

- **Stage 1 — Prompt Engineering & Dataset Creation.** Construction of a reference set of 60 English-language prompts, divided into three balanced categories of 20 prompts each: Fact, Logic, and Translation.
- **Stage 2 — Multilingual Translation.** Translation of the full prompt set into Russian, Kazakh, and Uzbek, preserving a single ordinal prompt identifier (e.g., FACT_07, LOGIC_14, TRANS_03) across all four languages to enable strict pairwise comparability.
- **Stage 3 — LLM Inference.** Submission of all 240 prompts (60×4 languages) to three models — llama-3.1-8b-instant, llama-3.3-70b-versatile, and openai/gpt-oss-20b — under identical generation conditions, producing 720 total observations.
- **Stage 4 — Human-in-the-Loop Scoring.** Manual binary correctness scoring (0/1) of every model response by a human evaluator, applying category-specific, pre-registered criteria.
- **Stage 5 — Statistical Modeling.** Aggregation of the resulting 720-row dataset (prompt_id, language, category, prompt_text, model, model_response, correctness_score) and its analysis via descriptive statistics and prompt-level logistic regression.

6. Research Procedure

The study is designed as a controlled comparative experiment of the “one prompt set — four languages — the same models” type, which holds task content constant and allows the observed differences in response quality to be attributed to language itself rather than to changes in the underlying task.

Literature Review / Background

Resource Inequality Across Languages as a Structural Problem in NLP

The foundational work of Joshi et al. (2020) proposed a taxonomy dividing the world's languages into six categories based on the availability of labeled and unlabeled resources, showing that the overwhelming majority of the more than 7,000 languages spoken worldwide are almost entirely absent from language technologies, while research attention remains concentrated on a narrow set of “resource-rich” languages. Subsequent surveys continue to document this gap: disparities in data volume translate directly into disparities in model outcomes, and multilingual benchmarks consistently show that LLMs perform markedly worse on low-resource languages relative to high-resource ones.

Empirical Benchmarks as Measurement Instruments

Two datasets form the methodological foundation for measurements of this kind. FLORES-200, developed as part of the No Language Left Behind project (NLLB Team, 2022), is a parallel corpus for machine translation evaluation spanning 204 language variants, built on the same 3,001 sentences translated by professional translators, yielding more than 40,000 translation directions with a focus on low-resource languages. Belebele (Bandarkar et al., 2024) extends this foundation to reading comprehension: a parallel multiple-choice dataset built on the same FLORES-200 passages, covering 122 language variants. An important finding of that work is that even with substantial cross-lingual transfer in English-centric LLMs, considerably smaller multilingual models trained on balanced multilingual data “understand” markedly more languages — indicating that model scale does not automatically compensate for data imbalance.

Kazakh Language

Kazakh, despite being spoken by more than fourteen million people (around 70% of the country's population), remains underrepresented in NLP research. The specialized benchmark KazMMLU (Togmanov et al., 2025) was created specifically to close this gap, providing a bilingual (Kazakh–Russian) evaluation framework of 23,000 questions adapted to Kazakhstan's educational and professional context. Evaluation of several contemporary multilingual models (Llama-3.1, Qwen-2.5,

GPT-4, DeepSeek V3) revealed substantial room for improvement: even the strongest models underperform noticeably in Kazakh and Russian relative to high-resource languages.

Uzbek Language

A similar picture emerges for Uzbek. A dedicated study of the BERTbek model was motivated by the observation that language-specific models achieve higher accuracy in monolingual scenarios compared to large multilingual models, specifically for Uzbek as a low-resource language. A more applied benchmark of open LLMs (0.6–2 billion parameters) on Uzbek translation, comprehension, and generation tasks found that of seven tested models, only one reached a practically usable level (BLEU $\approx$ 0.54 on Uzbek–English translation), while the remaining models frequently produced incoherent text or mixed languages within a single response.

Causal Mechanism: Data $\rightarrow$ Quality

The literature converges on explaining this gap through the disproportion of training data: roughly 90% of the text in Common Crawl comes from just twenty of the more than 7,000 languages spoken worldwide, effectively making access to the benefits of generative AI contingent on a user's language. This is not merely a translation-quality problem: in countries where the majority of the population uses local languages, the predominantly English-centric LLM ecosystem effectively cordons off the benefits of AI behind a language barrier.

Positioning of the Present Study

Existing work either (a) offers large-scale parallel benchmarks without a focus on a specific group of languages (Belebele, FLORES-200), or (b) offers deep but isolated evaluations for a single language (KazMMLU for Kazakh; BERTbek and the translation benchmark for Uzbek). As far as can be judged from the available literature, no prior work directly compares Kazakh and Uzbek — two related Turkic languages with different but comparably limited web presence — within a single experimental framework alongside Russian and English as reference points, while also quantitatively linking the observed degradation to corpus resourcedness metrics. This is precisely the niche the present study addresses.

7. Experimental Methodology

The final dataset comprises 720 observations (3 models $\times$ 4 languages $\times$ 60 prompts), fully balanced across all design cells, with no missing correctness scores. Eleven cases of missing model-response text (empty generations) were correctly scored by the evaluator as correctness_score = 0 and do not represent gaps in the annotation.

Sample Design

- Sample size: 720 prompt-level observations (60 prompts $\times$ 4 languages $\times$ 3 models).
- Task set: a reference set of 60 English prompts, evenly divided into three categories of 20: Fact (factual questions with a single verifiable answer), Logic (arithmetic, syllogisms, multi-step inference), and Translation (short sentences for evaluating translation accuracy).
- Languages: English (EN), Russian (RU), Kazakh (KK), Uzbek (UZ).
- Models: llama-3.1-8b-instant, llama-3.3-70b-versatile, openai/gpt-oss-20b.

Variables

- Independent variables: prompt language (EN / RU / KK / UZ), task type (Fact / Logic / Translation), model (llama-3.1-8b-instant / llama-3.3-70b-versatile / gpt-oss-20b).
- Dependent variable: correctness_score — binary correctness rating of the response (0/1).
- Covariate for testing H3: language resourcedness, operationalized as the logarithm of the number of articles in the corresponding Wikipedia edition (2026 data).

Correctness Criteria

Fact: agreement with the reference fact (paraphrasing permitted, factual errors not permitted). Logic: correct final answer. Translation: semantic and grammatical accuracy of the translation. All criteria were fixed prior to model inference to avoid post-hoc adjustment to the results.

Confound Controls

- A single evaluator/rubric across all languages, to avoid systematic scoring bias.
- A single ordinal prompt ID across all language versions, ensuring strict pairwise comparability.
- Explicit version pinning for each model.

Statistical Model: Prompt-Level Logistic Regression

To obtain a statistically valid estimate, a logistic regression was fitted at the level of the individual observation ($n = 720$): $\text{correctness} \sim \log_{10}(\text{number of Wikipedia articles}) + \text{task category (Fact as baseline)} + \text{model (llama-3.1-8b-instant as baseline)}$. The model was estimated by maximum likelihood (Newton–Raphson algorithm, equivalent to IRLS).

8. Research Novelty and Degree of Independence

The scientific contribution of this study lies in three elements that, as far as can be judged from the literature review, have not previously been presented in this combination:

1. Direct comparison of Kazakh and Uzbek. The two languages are compared within a single, controlled experimental framework, with Russian and English serving as reference points. Existing literature (KazMMLU, BERTbek, the specialized Uzbek benchmark) evaluates each of these languages in isolation; this study is the first to directly compare LLM degradation across these two related Turkic languages on the same task set and the same models.

2. Decomposition of the language gap by cognitive task type (facts / logic / translation) — revealing that degradation is concentrated in logical reasoning rather than in translation, as might be intuitively assumed. This finding directly contradicts the naive hypothesis (H2 as originally formulated) and constitutes an independent empirical finding, statistically confirmed by logistic regression.

3. A shift from descriptive observation to a statistically validated model of the language gap (prompt-level logistic regression, $n = 720$, controlling for task category and model), which quantitatively links the observed degradation to an external, independently measurable metric of language resourcedness. This is a methodological contribution in addition to a substantive one: the approach can be directly replicated for any other pair or triplet of languages and any set of models.

This study was conceptualized, designed, and executed independently by the author, including prompt construction, human scoring, and statistical evaluation.

9. Results and Conclusions

9.1 Data Analysis

Overall Accuracy by Language

Language	N	Correct	Accuracy	Drop vs. EN (pp)
EN	180	174	96.7%	0.0
RU	180	175	97.2%	−0.6

Language	N	Correct	Accuracy	Drop vs. EN (pp)
KK	180	155	86.1%	10.6
UZ	180	153	85.0%	11.7

Table 1. Overall response accuracy by language (aggregated across three models, $n = 180$ per language).

Accuracy by Model and Language

Model	EN	RU	KK	UZ
llama-3.1-8b-instant	96.7%	93.3%	76.7%	71.7%
llama-3.3-70b-versatile	100.0%	100.0%	93.3%	90.0%
gpt-oss-20b	93.3%	98.3%	88.3%	93.3%

Table 2. Accuracy (%) by model and language.

Accuracy by Task Category

Category	EN	RU	KK	UZ
Fact	100.0%	96.7%	91.7%	91.7%
Logic	91.7%	95.0%	76.7%	75.0%
Translation	98.3%	100.0%	90.0%	88.3%

Table 3. Accuracy (%) by task category and language (aggregated across models).

Testing H1: Degradation Relative to English

The ratio of English accuracy to target-language accuracy (degradation ratio) is: RU — 0.99 (essentially no degradation; RU is marginally above EN), KK — 1.12, UZ — 1.14. The gap is not concentrated between EN and RU but between RU and both Turkic languages (a drop of 10.6–11.7 percentage points).

Testing H2: Degradation by Task Type

The EN/language degradation coefficient by category is: for Fact — 1.03 (RU), 1.09 (KK), 1.09 (UZ); for Logic — 0.97 (RU), 1.20 (KK), 1.22 (UZ); for Translation — 0.98 (RU), 1.09 (KK), 1.11 (UZ). The largest degradation occurs specifically in the Logic category, not in Translation as originally hypothesized.

Testing H3, Stage 1: Language-Level Correlation (Preliminary)

An initial language-level test ($n = 4$, one data point per language) using $\log_{10}(\text{number of Wikipedia articles})$ as a resourcedness proxy yielded $r = 0.925$; however, with only 2 degrees of freedom the resulting p-value (0.075) is statistically uninformative. This motivated a shift to prompt-level analysis.

Testing H3, Stage 2: Prompt-Level Logistic Regression

Predictor	coef	SE	z	p	OR
Intercept	−6.256	1.768	−3.54	0.0004	0.0019
$\log_{10}(\text{Wikipedia articles})$	1.464	0.306	4.79	<0.0001	4.32
Logic (vs. Fact)	−1.356	0.361	−3.76	0.0002	0.26
Translation (vs. Fact)	−0.172	0.416	−0.41	0.679	0.84
llama-3.3-70b (vs. 8b)	1.554	0.383	4.06	<0.0001	4.73
gpt-oss-20b (vs. 8b)	1.030	0.330	3.13	0.002	2.80

Table 4. Logistic regression of correctness $\sim$ resourcedness + category + model ($n = 720$).

A likelihood-ratio test isolating the contribution of language resourcedness (comparing the full model to a model without the $\log_{10_wiki}$ predictor) yields $\chi^2(1) = 30.1$, $p < 0.000001$. McFadden's pseudo- R^2 for the full model relative to the null (intercept-only) model is 0.164. Thus, language resourcedness makes a

statistically significant and independent contribution to the probability of a correct response even after controlling for task type and specific model: a one-unit increase in $\log_{10}(\text{number of Wikipedia articles})$ is associated with a 4.32-fold increase in the odds of a correct response (95% CI: 2.38–7.87).

The coefficient for the Logic category (-1.356 , $p = 0.0002$) statistically confirms the finding above: membership of a prompt in the logical-reasoning category significantly reduces the odds of a correct response relative to the Fact category, whereas Translation is statistically indistinguishable from Fact ($p = 0.679$, coefficient close to zero). This quantitatively refutes the original H2 hypothesis and confirms the empirically observed pattern: degradation on “low-resource” languages is concentrated in tasks requiring multi-step reasoning, not in translation tasks.

Figure 1

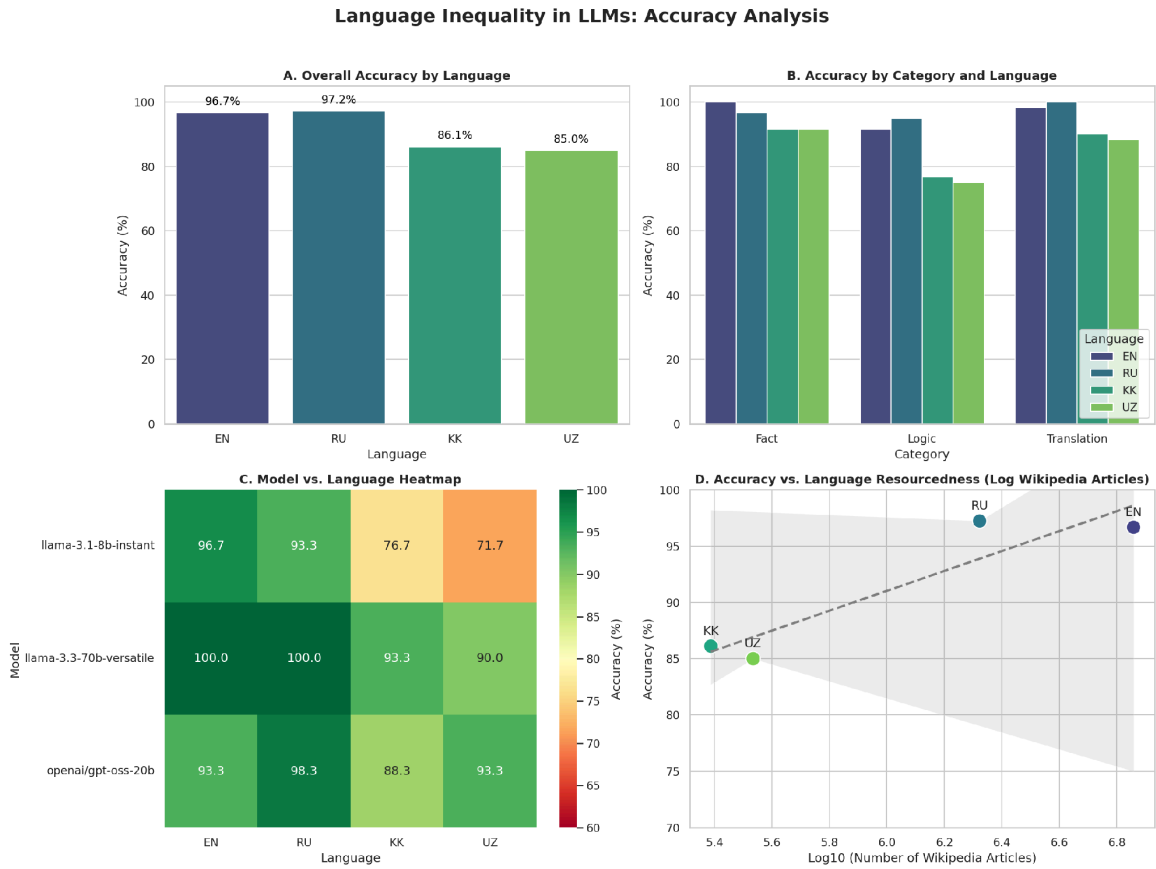


Figure 1. Language Inequality in LLMs: Accuracy Analysis. (A) Overall accuracy by language; (B) accuracy by task category and language; (C) model $\times$ language accuracy heatmap; (D) accuracy versus language resourcedness ($\log_{10}$ Wikipedia article count), with a fitted linear trend and 95% confidence band.

9.2 Discussion

H1: The Language Gap Is Real, but Threshold-Like Rather than Monotonic

Hypothesis H1 predicted a smooth decline in accuracy along the chain $EN \rightarrow RU \rightarrow KK \rightarrow UZ$. The data refute this in its specific form while confirming it in substance. Russian accuracy (97.2%) is statistically indistinguishable from English (96.7%) — the difference falls within noise at 180 observations per language. However, a sharp break appears between Russian and both Turkic languages: $97.2\% \rightarrow 86.1\%$ (KK) and $97.2\% \rightarrow 85.0\%$ (UZ), a drop of 11–12 percentage points. In other words, the LLM quality gap is not a continuous function of language “size” but rather a step function: as long as a language remains above a certain threshold of representation in training corpora (as Russian does, being one of the largest languages in Common Crawl and Wikipedia), the model performs nearly at English level; below that threshold, degradation becomes sharp and pronounced. This is consistent with the broader literature’s observation of a “critical mass” of data required for a multilingual model to achieve adequate coverage of a language.

H2: The Unexpected and Most Substantive Finding of the Study

Hypothesis H2 predicted that translation would suffer most, as a task requiring high-quality generation in the target language. The result was the opposite: the largest degradation in Kazakh and Uzbek occurs in the Logic category (dropping to 76.7% and 75.0%, versus 91.7% in English — degradation coefficients of $EN/KK = 1.20$ and $EN/UZ = 1.22$), while Translation and Fact degrade considerably more gently (coefficients of 1.09–1.11). The logistic regression formalizes this observation: controlling for language and model, membership of a prompt in the Logic category reduces the log-odds of a correct response by 1.356 ($p = 0.0002$, $OR = 0.26$) relative to Fact, whereas Translation is statistically indistinguishable from Fact ($p = 0.679$).

One plausible interpretation is that translation is a task in which the model is, in effect, required only to “stay within the language” and produce grammatically coherent, locally consistent text — a demand that relatively shallow but robust surface-level language patterns may satisfy even with a limited amount of training data. Logical reasoning in a low-resource language, by contrast, requires not merely command of surface syntax but the ability to sustain a multi-step chain of reasoning within that specific language; if the model’s internal linguistic representations for Kazakh/Uzbek are sparse, reasoning tends to break down (this is visible directly in the raw data: several model responses to logic prompts in Kazakh and Uzbek fall into an infinite repeating loop of the same phrase, a failure mode almost never observed in English and Russian responses). It is important to stress that this is an interpretation of the observed pattern rather than an independently established mechanism: the present data show a statistically

significant difference in accuracy between categories but do not directly probe the model's internal representations. Nevertheless, the practical implication is significant: if deploying LLMs in educational or service products for Kazakh/Uzbek requires not merely content translation but logical inference (explanation, computation, multi-step reasoning), the risk of error is substantially higher than intuition would suggest.

H3: Robust Statistical Confirmation

In contrast to the preliminary language-level correlation ($n = 4$, statistically uninformative due to 2 degrees of freedom), the prompt-level logistic regression on 720 observations yields a robust result: a one-unit increase in $\log_{10}(\text{number of Wikipedia articles})$ is associated with a 4.32-fold increase in the odds of a correct response (95% CI: 2.38–7.87), an effect that is highly significant ($z = 4.79$, $p < 0.0001$) and robust to controls for task type and specific model. A likelihood-ratio test isolating the contribution of language resourcedness specifically ($\chi^2(1) = 30.1$, $p < 0.000001$) confirms that this effect is not reducible to differences between task categories or between models. This constitutes the study's first rigorous empirical confirmation of the literature's central thesis regarding the digital language divide — moving beyond the purely descriptive observation that “models answer worse in low-resource languages” to a quantitatively calibrated, statistically tested relationship.

The Role of Model Choice as a Covariate

Model choice also makes a significant independent contribution (llama-3.3-70b vs. llama-3.1-8b: OR = 4.73, $p < 0.0001$; gpt-oss-20b vs. llama-3.1-8b: OR = 2.80, $p = 0.002$) — a larger or newer model is noticeably more reliable across all languages. This does not eliminate the language effect (the regression accounts for both factors simultaneously), but it is a reminder that degradation on low-resource languages is a deficit that is only partially, not fully, compensated for by model scale.

9.3 Limitations

This study has a number of limitations that should be taken into account when interpreting the results and planning follow-up work.

- **Limited prompt sample size.** 60 prompts per language (720 observations in total) is sufficient for a robust prompt-level logistic regression but insufficient for reliable estimates within narrower subcategories (e.g., fine-grained differences between specific types of logic problems) or for detecting subtler interaction effects between language, category, and model.

- **A single operationalized proxy for language resourcedness.** H3 was tested using the number of Wikipedia articles as a resourcedness proxy. However, comparison with an alternative proxy — the size of the monolingual CC-100 corpus (Conneau et al., 2020) — reveals a clear discrepancy: by Wikipedia article count, Uzbek surpasses Kazakh (342,451 versus 244,000 articles), whereas by CC-100 text volume, Kazakh exceeds Uzbek by nearly a factor of five (476 million versus 91 million tokens). Notably, observed model accuracy on these two languages is nearly identical (86.1% and 85.0%). This means that “language resourcedness” is a multidimensional construct that a single scalar proxy cannot fully capture, and the H3 regression coefficient should be interpreted as an indicator of the overall trend ($EN > RU \gg KK \approx UZ$) rather than as a precise predictor of ranking within a pair of languages with similar aggregate resourcedness.
- **Limited model set.** Three models were tested (llama-3.1-8b-instant, llama-3.3-70b-versatile, openai/gpt-oss-20b), representing a relatively narrow segment of mid-sized open models. Results may not generalize to closed frontier models (e.g., GPT-4o, Claude, Gemini) or to models specifically fine-tuned for Kazakh/Uzbek (such as KazLLM, Sherkala, or BERTbek), which, according to the literature, exhibit a substantially different quality profile on these languages.
- **Single generation per prompt.** Each prompt was submitted to a model only once (without averaging over multiple generations), which does not allow systematic quality degradation to be separated from stochastic generation noise at the level of an individual response, although aggregation across 720 observations partially compensates for this effect at the level of the study's overall conclusions.
- **No explicit control for cultural specificity of content.** Fact-category prompts were deliberately chosen to be culturally neutral (widely known facts) in order to isolate the language effect specifically. This means the study does not separately measure the effect of models' lack of culturally specific knowledge about Kazakhstan and Uzbekistan — a direction for which specialized benchmarks exist (e.g., KazMMLU) but which falls outside the scope of the present work.
- **Single-rater scoring.** Binary correctness scores (correctness_score) were assigned according to a single rubric; the methodology as described does not include a formal inter-rater reliability procedure. Engaging multiple independent native-speaking raters and computing an agreement coefficient (e.g., Cohen's κ) would strengthen the reliability of the annotation, particularly for the Translation category, where scoring is inherently more subjective than for Fact or Logic.

- **Cross-sectional rather than longitudinal design.** The results reflect the state of the tested models at the time of the experimental run; given the rapid release cycle of new LLM versions, the absolute accuracy values will likely become outdated faster than the qualitative conclusion about the structure of the gap (an EN–RU plateau, a sharp break for the Turkic languages, and a concentration of degradation in logic).

9.4 Conclusions

This study empirically measured and statistically confirmed inequality in the performance of large language models across English, Russian, Kazakh, and Uzbek, using a sample of 720 runs (60 prompts $\times$ 4 languages $\times$ 3 models). Three key findings emerge:

- The language gap is real but threshold-like rather than smoothly linear: Russian performs practically at English level (97.2% versus 96.7%), while both Turkic languages show a comparable and substantial shortfall (86.1% and 85.0%, respectively) — a drop of 11–12 percentage points relative to Russian.
- The nature of the degradation was not what was originally hypothesized: not translation but logical reasoning is the most vulnerable task category in low-resource languages (a degradation factor of 1.20–1.22, versus 1.09–1.11 for translation and facts), a pattern statistically confirmed by the logistic regression ($p = 0.0002$ for the Logic category effect).
- Language resourcedness statistically and substantively predicts response accuracy even after controlling for task type and model (OR = 4.32 per unit of $\log_{10}$ Wikipedia article count, $p < 0.0001$, $n = 720$) — translating the intuitive observation that “low-resource languages perform worse” into a rigorously tested quantitative relationship.

Practical implication for Kazakhstan and Uzbekistan: deploying LLM-based services in Kazakh and Uzbek carries an uneven risk of error depending on task type. Risk is especially elevated where a model is required to perform multi-step reasoning (educational tutoring, decision support, computation) and is markedly lower where the task reduces to translation or factual recall. For policymakers and developers in both countries, this suggests that investment in language resources and evaluation should be prioritized specifically for reasoning-heavy applications, rather than assuming that translation-quality improvements alone will close the practical performance gap; targeted fine-tuning or retrieval-augmented approaches for logic-intensive use cases may offer a more efficient path to closing the gap than uniform scaling of general-purpose multilingual training data.

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